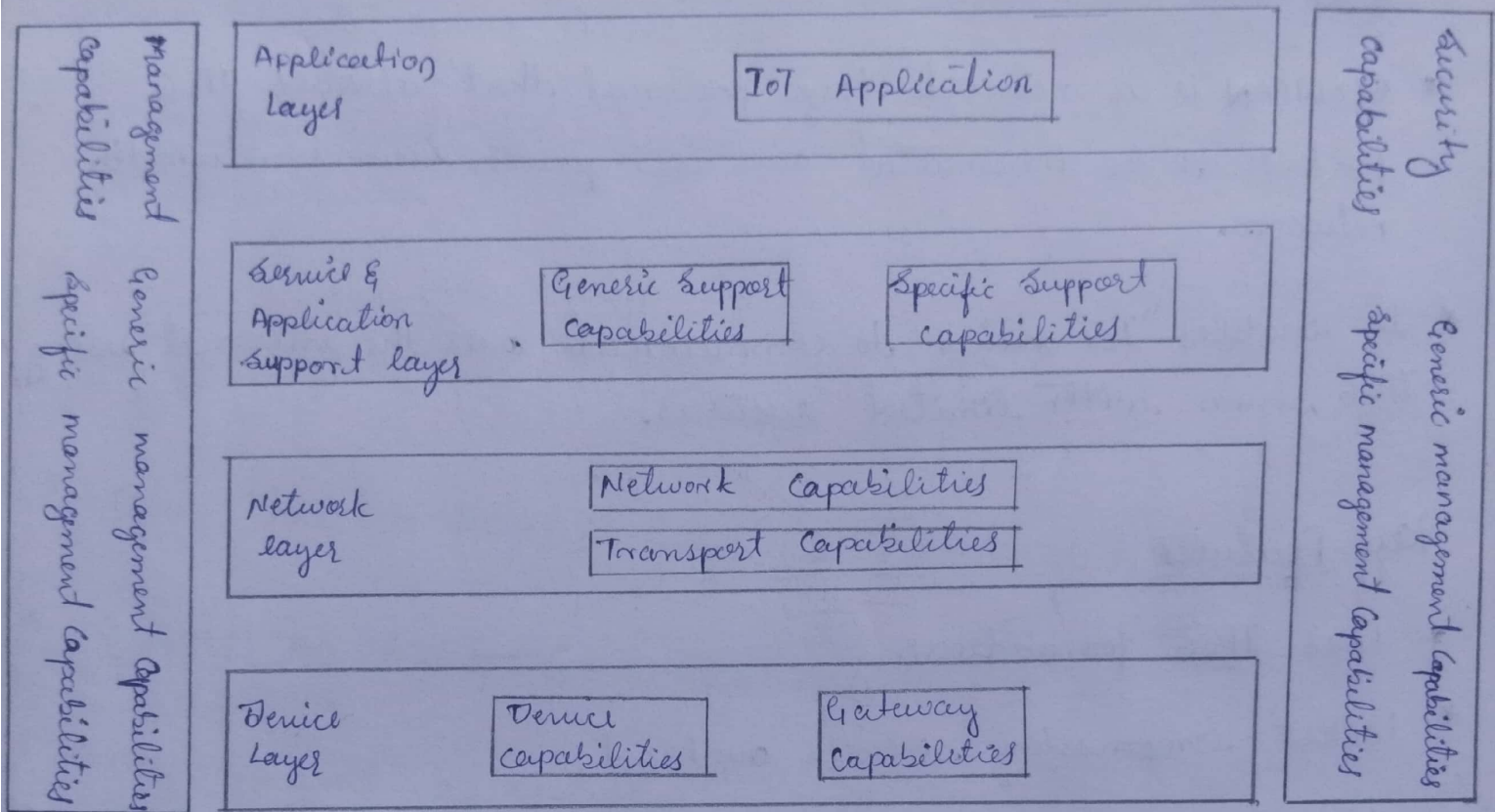


IOT ARCHITECTURE AND PROTOCOLS

Reference Model and architecture



Application layer: The user-facing layer that provides services, insights and control to end users.

Service and Application Support layer: provides the middleware and backend services to manage, process and store IoT data.

→ like cloud

Network layer: Handles the transmission of data between the devices layer and service layer.

Device layer: The foundation layer where IoT, devices, sensors and actuators reside.

IoT Protocols

2

ii) ~~6LoWPAN~~ 6LOWPAN:

- 6LOWPAN is a protocol that allows IPv6 over low-Power Wireless Personal Area Network used by small IoT devices communication
- * 6LOWPAN is a network layer protocol that allows IPv6 packets to be transmitted over low-power, low-bandwidth networks.
- * It enables IoT devices to communicate over the internet using IPv6, even with limited resources.

Key Features

- ✓ Uses IPv6 for addressing
- ✓ Packet compression reduces overhead.
- * Supports mesh networking
- ✓ Works over low-power wireless networks

Use Cases

- ✓ i) Smart Home: Smart thermostats, security cameras.
- ✓ ii) Industrial IoT: Wireless sensors in factories
- ✓ iii) Smart Cities: Streetlights, parking sensors

ii) RPL

(3)

- * Routing Protocol for low-Power and Lossy Networks.
- * RPL is a routing protocol designed for low-power, lossy networks (LLNs) like IoT sensor networks.
- * It is optimized for wireless communication where packet loss is common.

Key features

- * Uses Directed Acyclic Graph (DAG) to find best path.
- * Supports multi-hop Communication. (one node to another $A \rightarrow B \rightarrow C$)
- * Energy-efficient by reducing control message overhead.
- * work with IPv6 based IoT networks.

Use Cases

- i) Smart Agriculture: Soil moisture sensor networks.
- ii) Smart Grids: Power distribution monitoring.
- iii) Industrial IoT: Remote machine monitoring.

iii CoAP

- * Constrained Application Protocol
- * CoAP is a light-weight, application-layer protocol designed for resource-constrained IoT devices.
- * works on UDP (uses datagram protocol) instead of TCP (Transmission control protocol), reducing power and data overhead.

Key Features

- * RESTful architecture - Uses HTTP-like methods (GET, POST, PUT, DELETE).
- * Lightweight and low-power
- * Uses UDP instead TCP
- * Supports multicast Communication.

Use cases

- ~~i~~ Smart Lighting: Controlling streetlights remotely
- ~~ii~~ Environmental Monitoring: Collecting sensor data from remote locations.
- ~~iii~~ Home Automation: Controlling appliances via IoT hubs

iv) MQTT

Uses message queuing to transport ^{data} sensor

- Message Queuing Telemetry Transport.
- MQTT is a lightweight publish-subscribe messaging protocol designed for low-bandwidth, high-latency networks.
- uses a broker to manage communication between devices.

Key Features

- Publish-subscribe Model: Devices publish data to a topic and subscribers receive updates.
→ quality of service (in this case QoS=3)
- QoS Levels - Ensures reliable message delivery
- Low-bandwidth
- Supports TLS encryption for security
↳ Transport layer security.

Use Cases

- Smart home: Controlling IoT devices via mobile apps.
- Healthcare: Remote patients monitoring
- Industrial Automation: Machine-to-Machine (M2M) Communication.